



US012414370B2

(12) **United States Patent**
Choi

(10) **Patent No.:** **US 12,414,370 B2**

(45) **Date of Patent:** **Sep. 9, 2025**

(54) **DISPLAY APPARATUS USING SEMICONDUCTOR LIGHT-EMITTING DEVICES**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 100 days.

(21) Appl. No.: **17/436,262**

(22) PCT Filed: **Dec. 4, 2019**

(86) PCT No.: **PCT/KR2019/016978**

§ 371 (c)(1),

(2) Date: **Sep. 3, 2021**

(87) PCT Pub. No.: **WO2020/179989**

PCT Pub. Date: **Sep. 10, 2020**

(65) **Prior Publication Data**

US 2022/0165756 A1 May 26, 2022

(30) **Foreign Application Priority Data**

Mar. 4, 2019 (KR) 10-2019-0024916

(51) **Int. Cl.**

H10D 86/60 (2025.01)

H01L 23/00 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **H10D 86/60** (2025.01); **H01L 25/167** (2013.01); **H10D 86/441** (2025.01); **H01L 24/24** (2013.01);

(Continued)

(58) **Field of Classification Search**

CPC H01L 27/124; H01L 25/167; H01L 33/62; H01L 27/156; H01L 27/1214; H01L 29/786; H01L 33/42

See application file for complete search history.

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Primary Examiner — Yara B Green

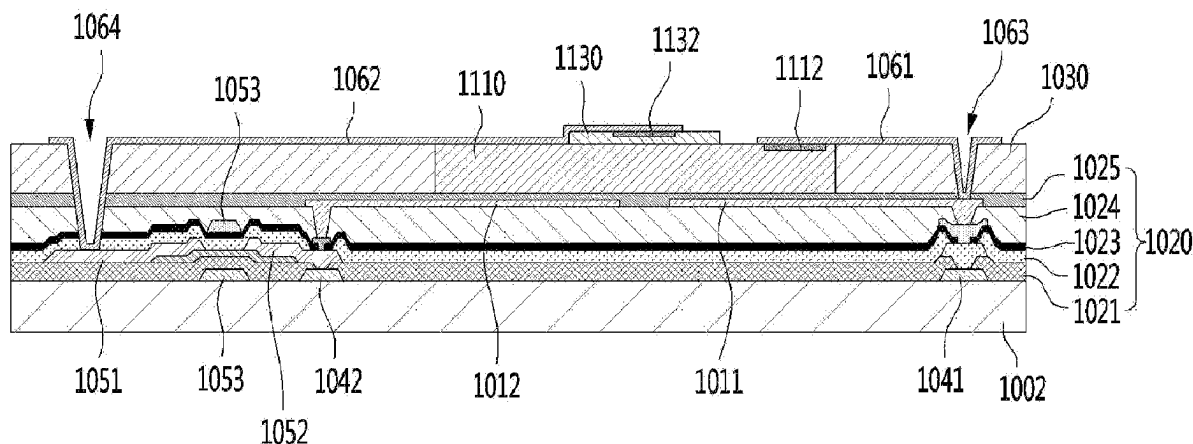
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(57) **ABSTRACT**

Discussed is a display device including: a substrate; a power wiring and a ground wiring disposed on the substrate and spaced apart from each other; a driving thin film transistor (TFT) disposed on the substrate and having a source terminal electrically connected to the ground wiring; at least one insulating layer disposed on the substrate; and a pair of assembly electrodes spaced apart from each other between the at least one insulating layer and the substrate, wherein the pair of assembly electrodes is configured to generate an electric field as a voltage is applied to any one of the pair of assembly electrodes.

22 Claims, 12 Drawing Sheets



- (51) **Int. Cl.**
H01L 25/16 (2023.01)
H10D 86/40 (2025.01)
- (52) **U.S. Cl.**
 CPC *H01L 24/95* (2013.01); *H01L 2224/24147*
 (2013.01); *H01L 2224/245* (2013.01); *H01L*
2224/95101 (2013.01); *H01L 2224/95136*
 (2013.01); *H01L 2224/95144* (2013.01); *H01L*
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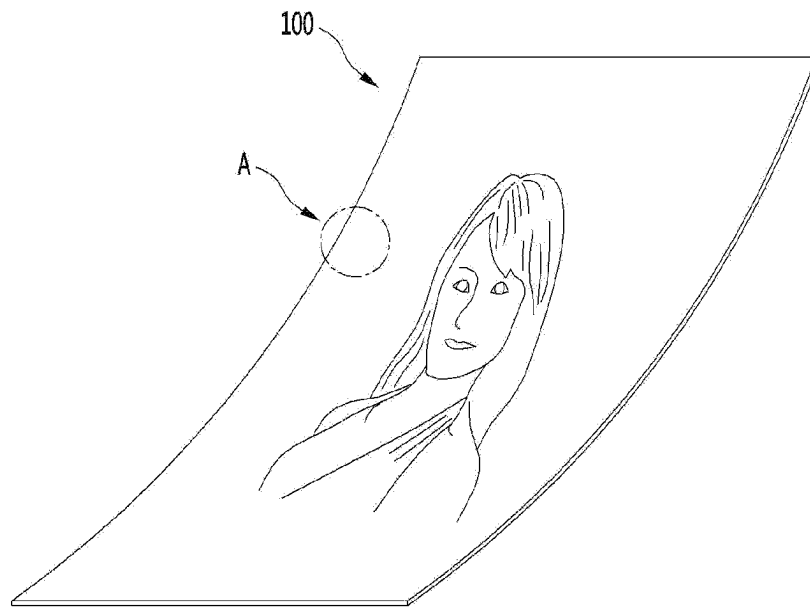
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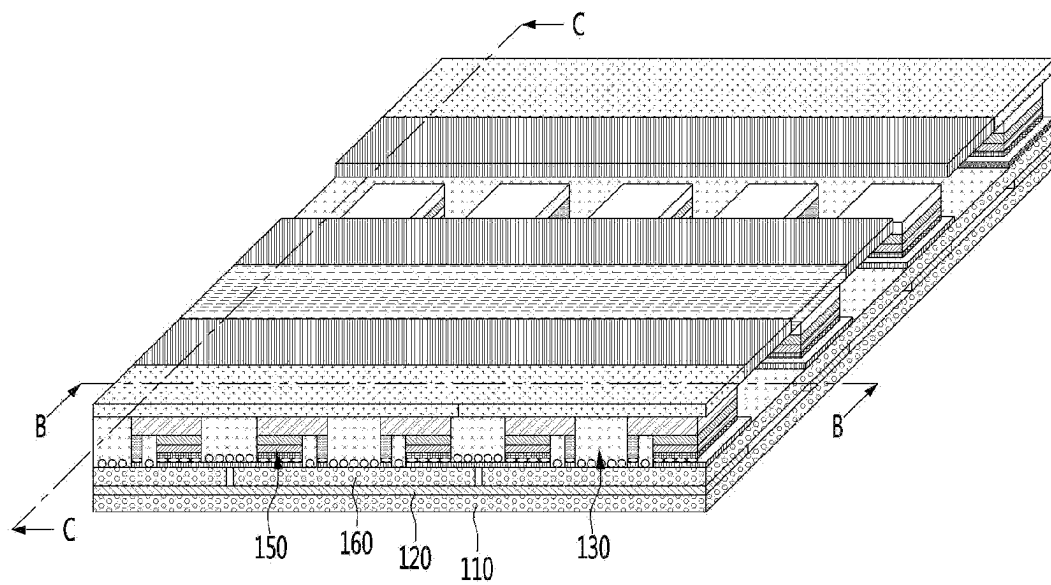
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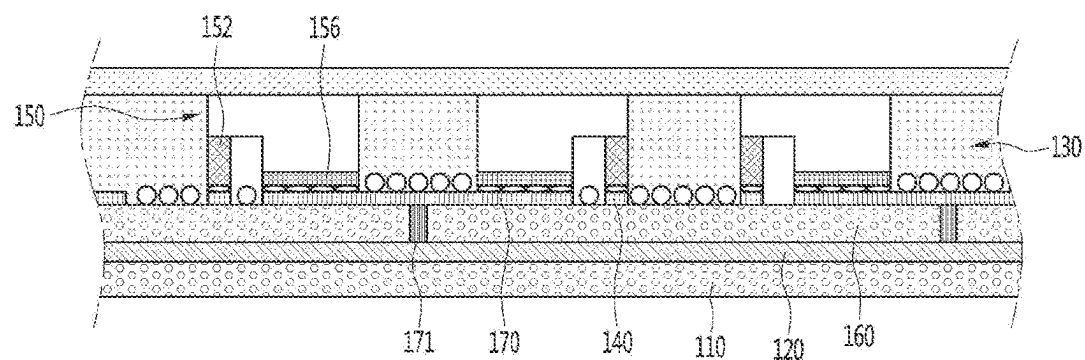
【FIG.1】



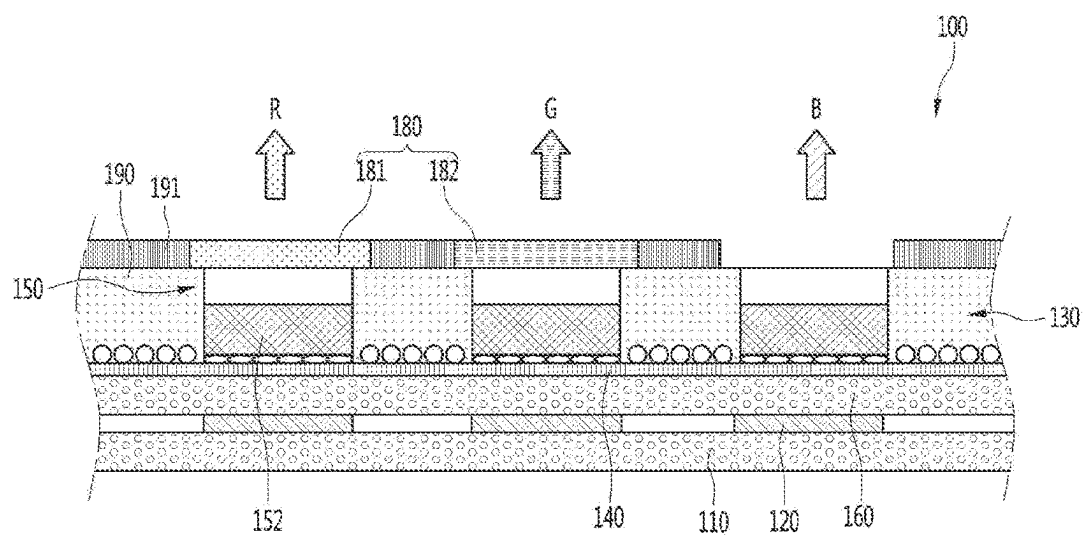
【FIG.2】



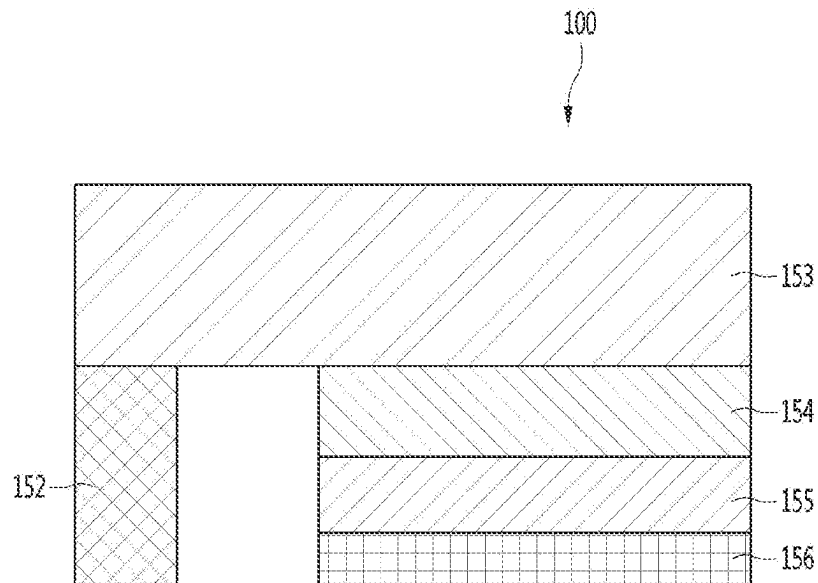
【FIG.3A】



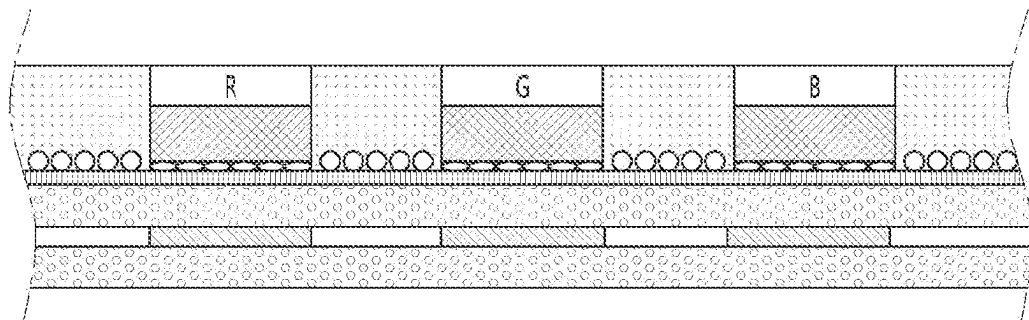
【FIG.3B】



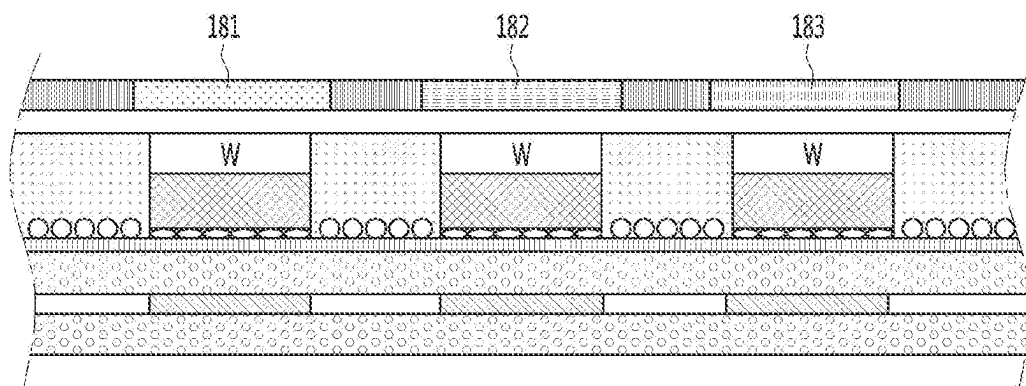
【FIG.4】



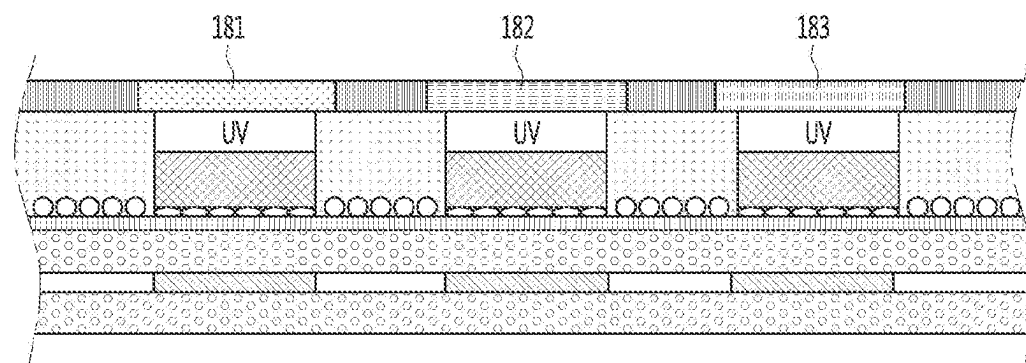
【FIG.5A】



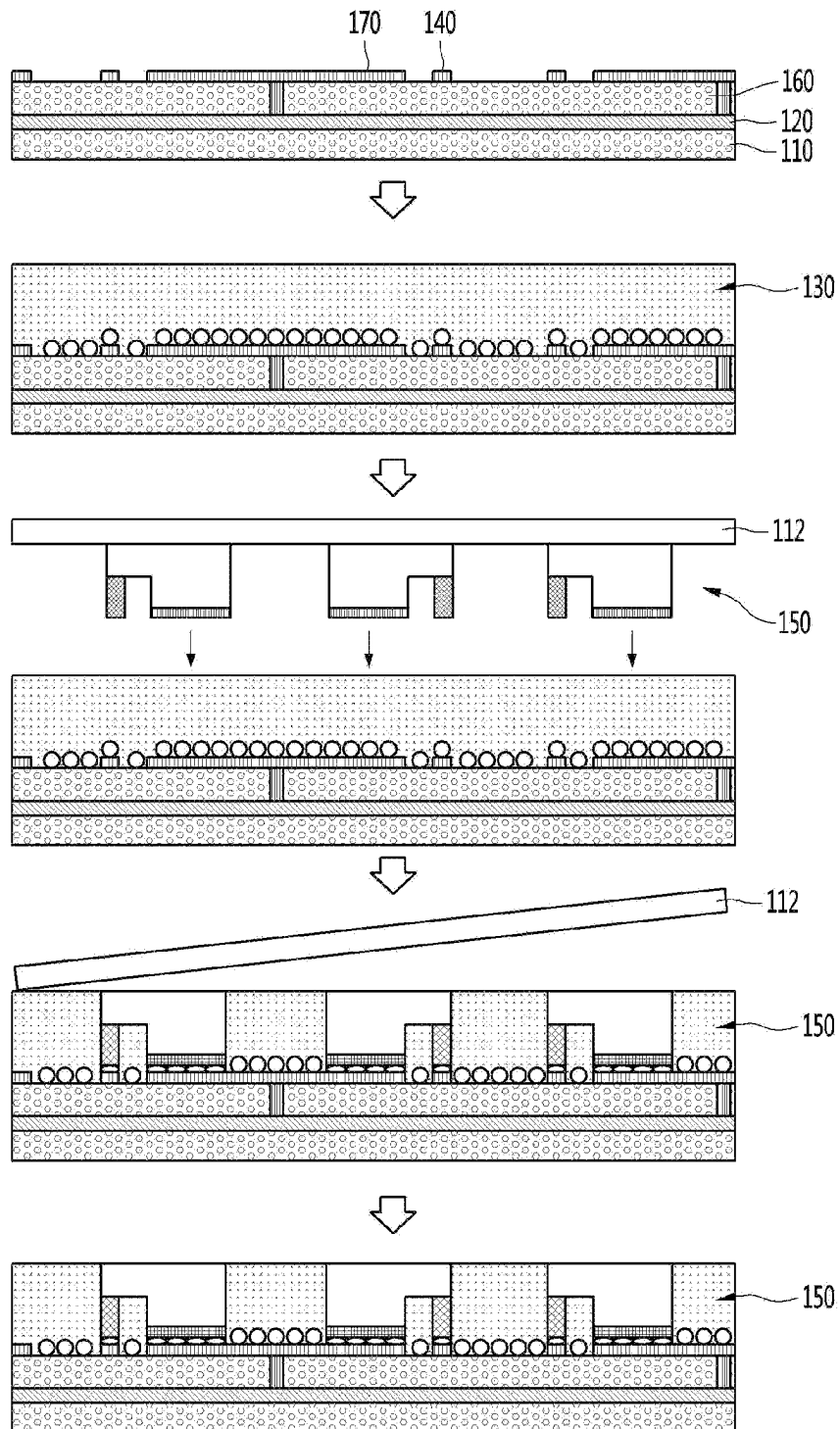
【FIG.5B】



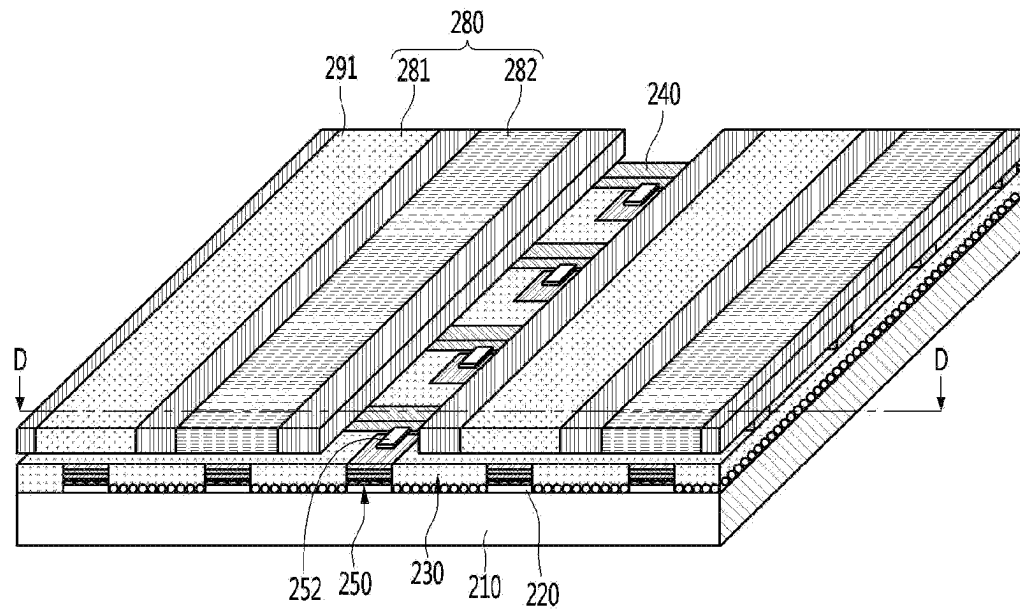
【FIG.5C】



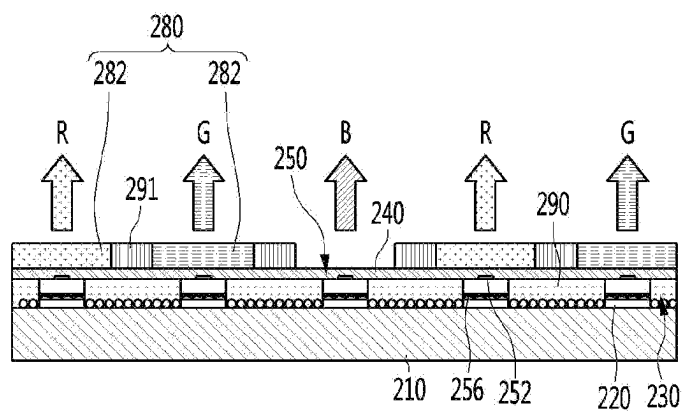
【FIG.6】



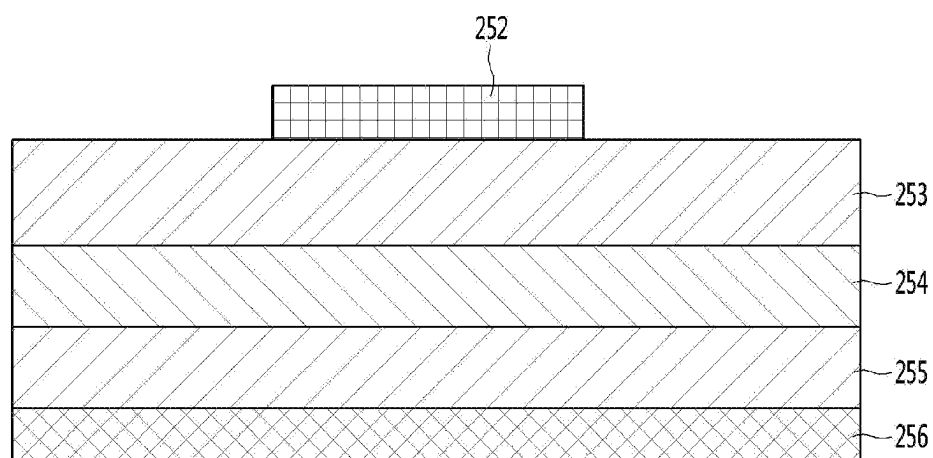
【FIG.7】



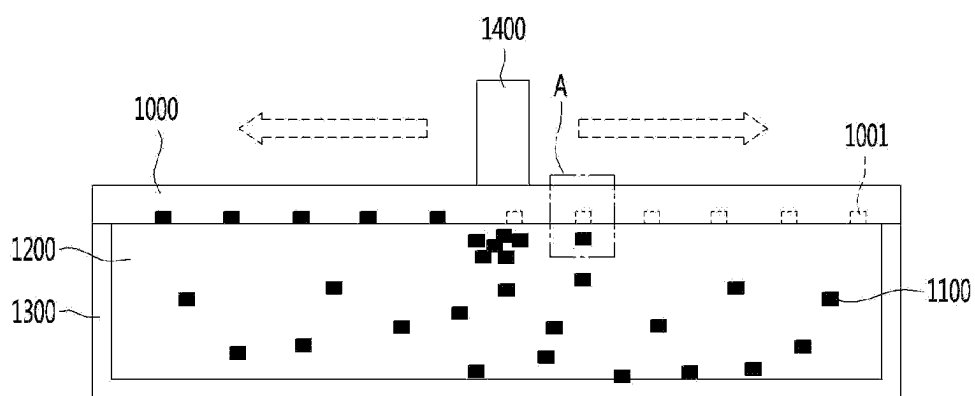
【FIG.8】



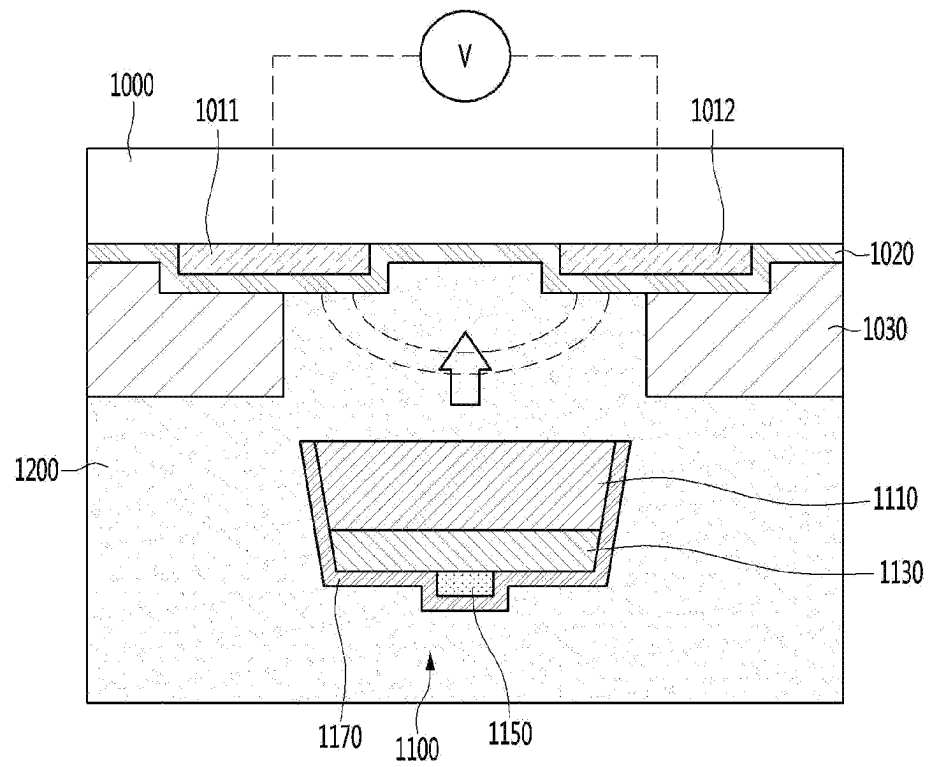
【FIG.9】



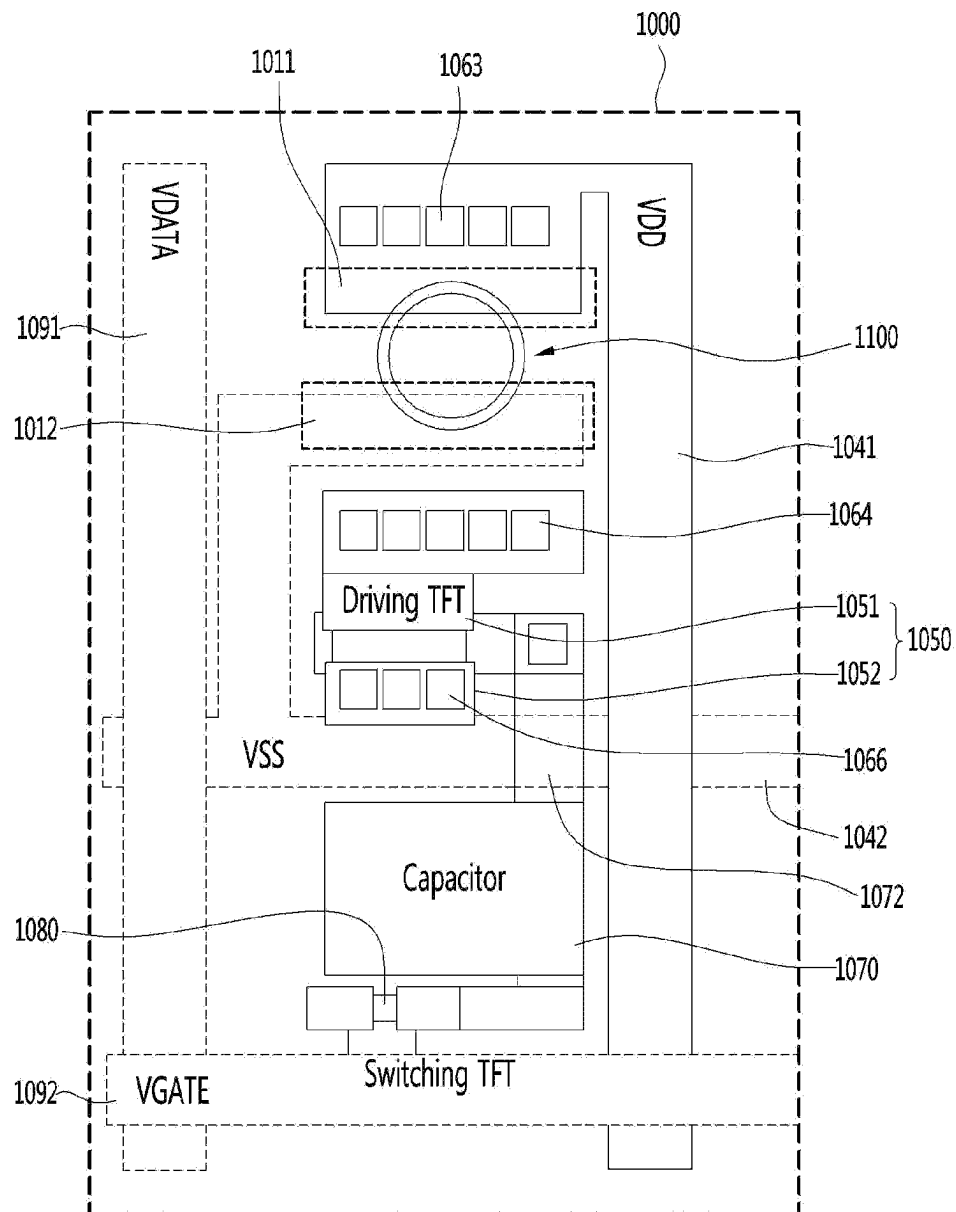
【FIG.10】



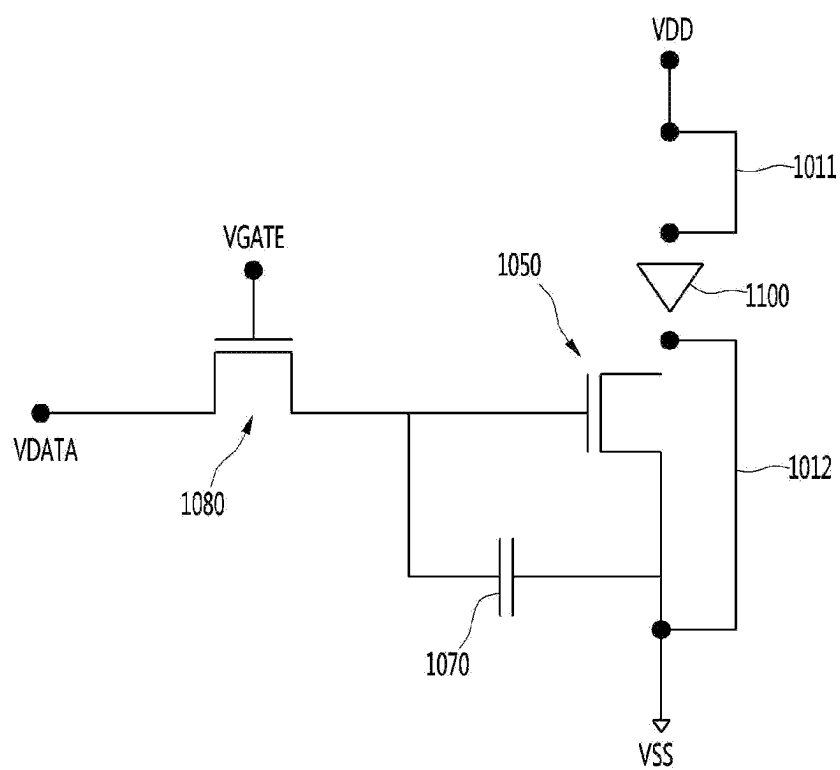
【FIG.11】



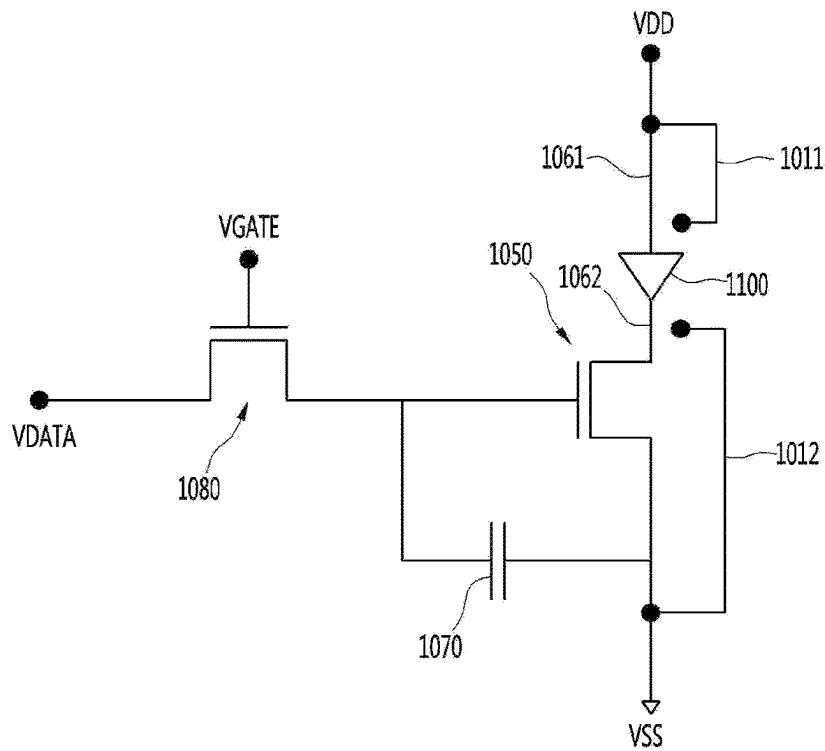
【FIG.12】



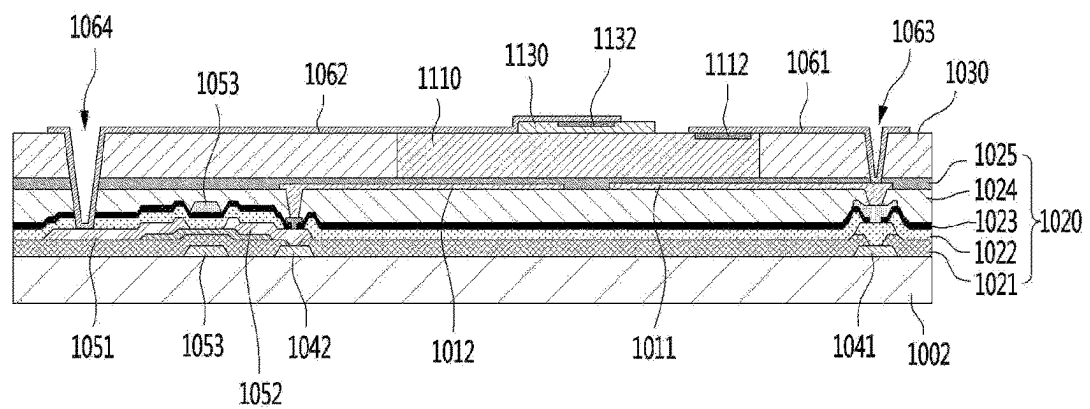
【FIG.13】

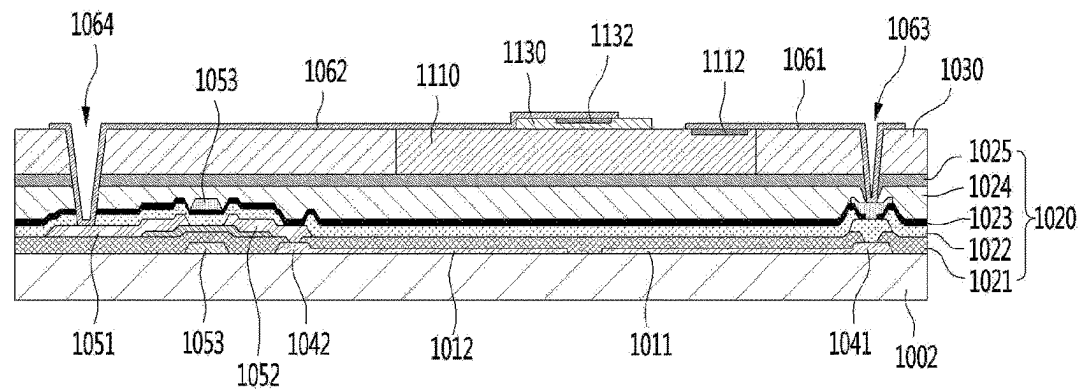


【FIG.14】



【FIG.15】





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DISPLAY APPARATUS USING SEMICONDUCTOR LIGHT-EMITTING DEVICES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2019/016978, filed on Dec. 4, 2019, which claims the benefit of earlier filing date and right of priority to Korean Application No. 10-2019-0024916, filed on Mar. 4, 2019 in the Republic of Korea, the entire contents of all these applications being hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD

The present invention relates to a display device using a semiconductor light emitting device.

BACKGROUND ART

Recently, in the field of display technology, display devices having excellent characteristics such as thinness and flexibility have been developed. On the other hand, currently commercialized main displays are represented by LCD (Liquid Crystal Display) and AMOLED (Active-Matrix Organic Light Emitting Diodes).

However, in the case of LCD, there are problems in that the response time is not fast and it is difficult to implement flexible, and in the case of AMOLED, there are weaknesses in that the life is short and the mass production yield is not good.

Meanwhile, a light emitting (diode (LED) is a well-known semiconductor light emitting device that converts electric current into light. It has been used as a light source for display images of electronic devices including information and communication devices. Accordingly, a method for solving the above problems by implementing a display using the semiconductor light emitting device may be proposed. Such a light emitting diode has various advantages, such as a long life, low power consumption, excellent initial driving characteristics, and high vibration resistance, compared to a filament-based light emitting device.

Meanwhile, in the case of a display using a semiconductor light emitting device, since the semiconductor light emitting device corresponding to each pixel must be coupled to the substrate during the transfer process, it may be relatively difficult to implement a large-screen high-pixel display. Accordingly, recently, a self-assembly method has been developed in which semiconductor light emitting devices injected into a fluid are moved to a substrate using an electromagnetic field and then assembled.

DISCLOSURE

Technical Problem

An object of the present invention is to provide a display device in which an efficient wiring process is implemented in an assembly electrode implemented in a display panel according to a self-assembly method.

Technical Solution

A display device according to an embodiment of the present invention may include a substrate, a power wiring

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and a ground wiring disposed on the substrate to be spaced apart from each other, a driving TFT disposed on the substrate and having a source terminal electrically connected to the ground wiring, at least one insulating layer disposed on the substrate and a pair of assembly electrodes spaced apart from each other between any one of the at least one insulating layer and the substrate. And the pair of assembly electrodes may generate an electric field as a voltage is applied to any one of the assembly electrodes.

A first assembly electrode of the pair of assembly electrodes can be electrically connected to the power wiring, and a second assembly electrode of the pair of assembly electrodes can be electrically connected to the ground wiring.

According to an embodiment, the pair of assembly electrodes can be formed on the substrate, the first assembly electrode can be formed to extend from the power wiring, and the second assembly electrode can be formed to extend from the ground wiring.

According to an embodiment, the at least one insulating layer may include a first insulating layer covering the substrate, the power wiring, and the ground wiring, the first assembly electrode, the second assembly electrode, and the source terminal can be formed on the first insulating layer, the first assembly electrode can be electrically connected to the power wiring through an etching region formed on the power wiring of the first insulating layer, and the second assembly electrode can be formed to extend from the source terminal and can be electrically connected to the ground wire through an etch region formed on the ground wire in the first insulating layer.

According to an embodiment, the at least one insulating layer may include a first insulating layer covering the substrate, the power wiring, and the ground wiring, a second insulating layer covering the source terminal formed on the first insulating layer and covering the first insulating layer and a third insulating layer covering the first assembly electrode and the second assembly electrode formed on the second insulating layer, and the second insulating layer. And the first assembly electrode may be electrically connected to the power wiring through an etch region formed in each of the first insulating layer and the second insulating layer on the power wiring and the second assembly electrode may be electrically connected to the source terminal and the ground wire through an etch region formed in the second insulating layer on the ground wire.

The display device may further include a semiconductor light emitting device assembled on the at least one insulating layer, and the semiconductor light emitting device can be positioned on the pair of assembly electrodes.

The display device may further include a partition wall formed on the at least one insulating layer, and a portion of the partition wall can be removed so that a bottom surface of the semiconductor light emitting device can be in contact with the at least one insulating layer.

The display device may further include a first wiring electrically connecting a first electrode formed in the semiconductor light emitting device and the power wiring and a second wiring electrically connecting a second electrode formed on the semiconductor light emitting device and a drain terminal of the driving TFT.

In an embodiment, the first wiring can be electrically connected to the power wiring through a partition wall on the power wiring and a first trench region formed by etching the at least one insulating layer and the second wiring can be electrically connected to the drain terminal through a partition wall on the drain terminal and a second trench region formed by etching the at least one insulating layer.

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According to an embodiment, each of the first wiring and the second wiring can be implemented as a transparent electrode.

According to an embodiment, the semiconductor light emitting device may include a magnetic layer having a magnetic material.

Advantageous Effects

According to an embodiment of the present invention, the assembly electrode for self-assembly of the semiconductor light emitting device may be connected to the power supply wiring and the ground wiring of the TFT driving circuit. Accordingly, a separate power wiring for applying a voltage to the assembly electrode may not be implemented in the panel. Therefore, the wiring structure of the panel is prevented from becoming complicated, noise due to wiring can be reduced, and a space utilization within the pixel can be prevented from being lowered.

DESCRIPTION OF DRAWINGS

FIG. 1 is a conceptual diagram illustrating an embodiment of a display device using a semiconductor light emitting device of the present invention.

FIG. 2 is a partially enlarged view of part A of FIG. 1, and FIGS. 3A and 3B are cross-sectional views taken along lines B-B and C-C of FIG. 2.

FIG. 4 is a conceptual diagram illustrating a flip-chip type semiconductor light emitting device of FIG. 3.

FIG. 5A to 5C are conceptual diagrams illustrating various forms of implementing colors in relation to a flip-chip type semiconductor light emitting device.

FIG. 6 is a cross-sectional view showing a method of manufacturing a display device using the semiconductor light emitting device of the present invention.

FIG. 7 is a perspective view showing another embodiment of a display device using the semiconductor light emitting device of the present invention.

FIG. 8 is a cross-sectional view taken along line D-D in FIG. 7.

FIG. 9 is a conceptual diagram illustrating the vertical semiconductor light emitting device of FIG. 8.

FIG. 10 is a diagram schematically illustrating an embodiment of a method for assembling a semiconductor light emitting device to a display panel by a self-assembly method.

FIG. 11 is an enlarged view of part A of FIG. 10.

FIG. 12 is a diagram schematically showing a circuit configuration of a display panel according to an embodiment of the present invention.

FIGS. 13 to 14 are circuit diagrams according to the circuit configuration of the display panel shown in FIG. 12.

FIGS. 15 to 18 are exemplary views showing various arrangement positions of the assembly electrode according to an embodiment of the present invention.

MODE FOR INVENTION

Hereinafter, the embodiments disclosed in the present specification will be described in detail with reference to the accompanying drawings, but the same or similar elements are given the same reference numerals regardless of the reference numerals, and the redundant description thereof will be omitted. The suffixes "module" and "part" for elements used in the following description are given or mixed in consideration of only the ease of writing the

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specification, and do not have distinct meanings or roles by themselves, in addition, in describing the embodiments disclosed in the present specification, if it is determined that detailed descriptions of related known technologies may obscure the meaning of the embodiments disclosed in the present specification, the detailed description thereof will be omitted. In addition, it should be noted that the accompanying drawings are only for easy understanding of the embodiments disclosed in the present specification, and should not be construed as limiting the technical spirit disclosed herein by the accompanying drawings.

Also, when an element, such as a layer, region, or substrate, is referred to as being "on" another element, it can be understood that it may be directly on the other element or intervening elements may be present in therebetween.

The display device described in this specification may include a mobile phone, a smart phone, a laptop computer, a digital broadcasting terminal, a personal digital assistant (PDA), a portable multimedia player (PMP), a navigation system, a slate PC, a Tablet PC, a Ultra Book, a digital TV, a desktop computer, and the like. However, it will be readily apparent to those skilled in the art that the configuration according to the embodiment described herein may be applied to a display capable device even in a new product form to be developed later.

FIG. 1 is a conceptual diagram illustrating an embodiment of a display device using a semiconductor light emitting device of the present invention.

As shown in drawings, information processed by the control unit of the display apparatus 100 may be displayed using a flexible display.

A flexible display may include a display that can be bent, crooked, twisted, folded, or rolled by an external force. For example, the flexible display may be a display manufactured on a thin and flexible substrate that can be bent, crooked, folded, or rolled like paper while maintaining the display characteristics of a conventional flat panel display.

In a state in which the flexible display is not bent (for example, a state having an infinite radius of curvature, hereinafter referred to as a first state), the display area of the flexible display becomes a flat surface. In a state bent by an external force in the first state (for example, a state having a finite radius of curvature, hereinafter referred to as a second state), the display area may be a curved surface. As illustrated, the information displayed in the second state may be visual information output on the curved surface. Such visual information is implemented by independently controlling the light emission of sub-pixels arranged in a matrix form. The unit pixel means a minimum unit for realizing one color.

The unit pixel of the flexible display may be implemented by a semiconductor light emitting device. In the present invention, a light emitting diode (LED) is exemplified as a type of semiconductor light emitting device that converts current into light. The light emitting diode is formed in a small size, so that it can serve as a unit pixel even in the second state.

Hereinafter, a flexible display implemented using the light emitting diode will be described in more detail with reference to the drawings.

FIG. 2 is a partially enlarged view of part A of FIG. 1, FIGS. 3A and 3B are cross-sectional views taken along lines B-B and C-C of FIG. 2, and FIG. 4 is a conceptual diagram of a flip-chip type semiconductor light emitting device of FIG. 3A, and FIGS. 5A to 5C are conceptual diagrams illustrating various forms of implementing colors in relation to a flip-chip type semiconductor light emitting device.

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Referring to FIGS. 2, 3A, and 3B, as the display device **100** using a semiconductor light emitting device, the display device **100** using a passive matrix (PM) type semiconductor light emitting device is exemplified. However, the example described below is also applicable to an active matrix (AM) type semiconductor light emitting device.

The display device **100** may include a substrate **110**, a first electrode **120**, a conductive adhesive layer **130**, a second electrode **140**, and a plurality of semiconductor light emitting devices **150**.

The substrate **110** may be a flexible substrate. For example, to implement a flexible display device, the substrate **110** may include glass or polyimide (PI). In addition, any material such as polyethylene naphthalate (PEN) or polyethylene terephthalate (PET) may be used as long as it has insulating properties and is flexible. In addition, the substrate **110** may be made of either a transparent material or an opaque material.

The substrate **110** may be a wiring substrate on which the first electrode **120** is disposed, and thus, the first electrode **120** may be located on the substrate **110**.

As shown in drawings, the insulating layer **160** may be disposed on the substrate **110** on which the first electrode **120** is positioned, and the auxiliary electrode **170** may be positioned on the insulating layer **160**. In this case, a state in which the insulating layer **160** is laminated on the substrate **110** may be a single wiring board. More specifically, the insulating layer **160** may be made of an insulating and flexible material such as polyimide (PI), PET, or PEN, and may be integrally formed with the substrate **110** to form a single substrate.

The auxiliary electrode **170** is an electrode that electrically connects the first electrode **120** and the semiconductor light emitting device **150**. The auxiliary electrode **170** may be disposed on the insulating layer **160** and may be disposed to correspond to the position of the first electrode **120**. For example, the auxiliary electrode **170** may have a dot shape and may be electrically connected to the first electrode **120** by an electrode hole **171** penetrating the insulating layer **160**. The electrode hole **171** may be formed by filling the via hole with a conductive material.

Referring to the drawings, the conductive adhesive layer **130** may be formed on one surface of the insulating layer **160**, but it is not limited thereto. For example, a layer performing a specific function is formed between the insulating layer **160** and the conductive adhesive layer **130**, or the conductive adhesive layer **130** can be disposed on the substrate **110** without the insulating layer **160**. In a structure in which the conductive adhesive layer **130** is disposed on the substrate **110**, the conductive adhesive layer **130** may serve as an insulating layer.

The conductive adhesive layer **130** may be a layer having adhesiveness and conductivity, and for this purpose, a material having conductivity and a material having adhesiveness may be mixed in the conductive adhesive layer **130**. In addition, the conductive adhesive layer **130** may have ductility, thereby enabling a flexible function in the display device.

For this example, the conductive adhesive layer **130** may be an anisotropic conductive film (ACF), an anisotropic conductive paste, a solution containing conductive particles, or the like. The conductive adhesive layer **130** may be configured as a layer that allows electrical interconnection in the Z direction passing through the thickness, but has electrical insulation in the horizontal X-Y direction. Accord-

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ingly, the conductive adhesive layer **130** may be referred to as a Z-axis conductive layer (hereinafter, referred to as a 'conductive adhesive layer').

The anisotropic conductive film may be a film in which an anisotropic conductive medium is mixed with an insulating base member, and when heat and pressure are applied, only a specific portion has conductivity by the anisotropic conductive medium. Hereinafter, it will be described that heat and pressure are applied to the anisotropic conductive film, but other methods are also possible in order for the anisotropic conductive film to have partial conductivity. In this method, for example, only one of the heat and pressure may be applied or UV curing may be performed.

In addition, the anisotropic conductive medium may be, for example, conductive balls or conductive particles. As shown, in this example, the anisotropic conductive film is a film in which conductive balls are mixed with an insulating base member, and when heat and pressure are applied, only a specific portion has conductivity by the conductive balls. The anisotropic conductive film may be in a state in which the core of the conductive material contains a plurality of particles covered by an insulating film made of a polymer material. At this time, the shape of the core may be deformed to form a layer in contact with each other in the thickness direction of the film. As a more specific example, heat and pressure are applied as a whole to the anisotropic conductive film, and electrical connection in the Z-axis direction is partially formed due to a height difference of an object adhered by the anisotropic conductive film.

As another example, the anisotropic conductive film may be in a state in which an insulating core contains a plurality of particles coated with a conductive material. In this case, the conductive material is deformed (pressed) in the portion to which heat and pressure are applied, so that it has conductivity in the thickness direction of the film. As another example, a form in which the conductive material penetrates the insulating base member in the Z-axis direction to have conductivity in the thickness direction of the film is also possible. In this case, the conductive material may have a pointed end.

As shown in drawings, the anisotropic conductive film may be a fixed array anisotropic conductive film (ACF) in which conductive balls are inserted into one surface of the insulating base member. More specifically, the insulating base member is formed of a material having an adhesive property, and the conductive balls are intensively disposed on the bottom of the insulating base member, and when heat and pressure are applied from the base member, it is deformed together with the conductive balls and has conductivity in the vertical direction.

However, the present invention is not necessarily limited thereto, and the anisotropic conductive film may have a form in which conductive balls are randomly mixed in an insulating base member. Or it can be composed of a plurality of layers and conductive balls are arranged on any one layer double-ACF), etc. are all possible.

The anisotropic conductive paste may be in the form of a paste and a conductive ball combined such that it may be a paste in which conductive balls are mixed with an insulating and adhesive base material. Also, a solution containing conductive particles may be a solution containing conductive particles or nano particles.

Referring back to the drawings, the second electrode **140** may be spaced apart from the auxiliary electrode **170** and may be located in the insulating layer **160**. That is, the conductive adhesive layer **130** can be disposed on the

insulating layer **160** in which the auxiliary electrode **170** and the second electrode **140** are located.

A conductive adhesive layer **130** can be formed in a state in which the auxiliary electrode **170** and the second electrode **140** are positioned on the insulating layer **160**. Next, heat and pressure are applied to the semiconductor light emitting device **150** to connect the semiconductor light emitting device **150** in a flip-chip form, so that the semiconductor light emitting device **150** can be electrically connected to the first electrode **120** and the second electrode **140**.

Referring to FIG. **4**, the semiconductor light emitting device may be a flip chip type light emitting device.

For example, the semiconductor light emitting device may include a p-type electrode **156**, a p-type semiconductor layer **155** on which the p-type electrode **156** is formed, and an active layer **154** formed on the p-type semiconductor layer **155**, an n-type semiconductor layer **153** formed on the active layer **154** and an n-type electrode **152** spaced apart from the p-type electrode **156** in the horizontal direction on the n-type semiconductor layer **153**. In this case, the p-type electrode **156** may be electrically connected to the auxiliary electrode **170** and the conductive adhesive layer **130**, and the n-type electrode **152** may be electrically connected to the second electrode **140**.

Referring back to FIGS. **2**, **3A**, and **3B**, the auxiliary electrode **170** may be formed to be elongated in one direction, so that one auxiliary electrode can be electrically connected to the plurality of semiconductor light emitting devices **150**. For example, p-type electrodes of left and right semiconductor light emitting devices with the auxiliary electrode as the center may be electrically connected to one auxiliary electrode.

More specifically, the semiconductor light emitting device **150** may be press-fitted into the conductive adhesive layer **130** by heat and pressure. Accordingly, only the portion between the p-type electrode **156** and the auxiliary electrode **170** of the semiconductor light emitting device **150** and the portion between the n-type electrode **152** and the second electrode **140** of the semiconductor light emitting device **150** can have conductivity. The remaining portion without press-fitting of the semiconductor light emitting device does not have conductivity. As described above, the conductive adhesive layer **130** not only interconnects the semiconductor light emitting device **150** with the auxiliary electrode **170**, the second electrode **140**, but also forms an electrical connection.

In addition, the plurality of semiconductor light emitting devices **150** constitute a light emitting device array, and the phosphor layer **180** can be formed on the light emitting device array.

The light emitting device array may include a plurality of semiconductor light emitting devices having different luminance values. Each semiconductor light emitting device **150** constitutes a unit pixel and can be electrically connected to the first electrode **120**. For example, there may be a plurality of first electrodes **120**, the semiconductor light emitting devices may be arranged in, for example, several columns, and the semiconductor light emitting devices of each column may be electrically connected to any one of the plurality of first electrodes.

In addition, since the semiconductor light emitting devices are connected in a flip-chip form, semiconductor light emitting devices grown on a transparent dielectric substrate can be used. In addition, the semiconductor light emitting devices may be, for example, nitride semiconductor light emitting devices. Since the semiconductor light emit-

ting device **150** has excellent luminance, individual unit pixels can be configured even with a small size.

As shown in drawings, the partition walls **190** may be formed between the semiconductor light emitting devices **150**. In this case, the partition wall **190** may serve to separate individual unit pixels from each other, and may be integrally formed with the conductive adhesive layer **130**. For example, by inserting the semiconductor light emitting device **150** into the anisotropic conductive film, the base member of the anisotropic conductive film may form the partition wall.

In addition, when the base member of the anisotropic conductive film is black, the partition wall **190** may have reflective properties and increase contrast even without a separate black insulator.

As another example, a reflective partition wall may be separately provided as the partition wall **190**. In this case, the partition wall **190** may include a black or white insulator depending on the purpose of the display device. When the partition wall of the white insulator is used, it is possible to increase reflectivity, and when the partition wall of the black insulator is used, it is possible to have reflective properties and increase the contrast.

A phosphor layer **180** may be located on an outer surface of the semiconductor light emitting device **150**. For example, the semiconductor light emitting device **150** is a blue semiconductor light emitting device that emits blue (B) light, and the phosphor layer **180** functions to convert the blue (B) light into the color of a unit pixel. The phosphor layer **180** may be a red phosphor **181** or a green phosphor **182** constituting an individual pixel.

That is, a red phosphor **181** capable of converting blue light into red (R) light may be stacked on the blue semiconductor light emitting device at a position constituting the red unit pixel. In addition, a green phosphor **182** capable of converting blue light into green (G) light may be stacked on the blue semiconductor light emitting device at a position forming the green unit pixel. In addition, only the blue semiconductor light emitting device may be used alone in the portion constituting the blue unit pixel. In this case, unit pixels of red (R), green (G), and blue (B) may form one pixel. More specifically, a phosphor of one color may be stacked along each line of the first electrode **120**. Accordingly, one line in the first electrode **120** may be an electrode for controlling one color. That is, red (R), green (G), and blue (B) may be sequentially disposed along the second electrode **140**, thereby realizing a unit pixel.

However, the present invention is not necessarily limited thereto, and instead of the phosphor, the semiconductor light emitting device **150** and the quantum dot (QD) are combined to implement red (R), green (G) and blue (B) unit pixels.

In addition, a black matrix **191** may be disposed between the respective phosphor layers to improve contrast. That is, the black matrix **191** may improve contrast of light and dark.

However, the present invention is not necessarily limited thereto, and other structures for implementing blue, red, and green colors may be applied.

Referring to FIG. **5A**, each semiconductor light emitting device **150** is mainly gallium nitride (GaN), indium (In) and/or aluminum (Al) can be added together to emit various light including blue such that can be implemented as a high-output light emitting device.

In this case, the semiconductor light emitting device **150** may be a red, green, and blue semiconductor light emitting device to form a sub-pixel, respectively. For example, red, green, and blue semiconductor light emitting devices R, G, and B are alternately arranged, and unit pixels of red, green,

and blue are formed by the red, green and blue semiconductor light emitting devices such that a full-color display can be realized.

Referring to FIG. 5B, the semiconductor light emitting device may include a white light emitting device W in which a yellow phosphor layer is provided for each device. In this case, a red phosphor layer **181**, a green phosphor layer **182**, and a blue phosphor layer **183** may be provided on the white light emitting device W to form a unit pixel. In addition, a unit pixel may be formed on the white light emitting device W by using a color filter in which red, green, and blue are repeated.

Referring to FIG. 5C, a structure in which a red phosphor layer **181**, a green phosphor layer **182**, and a blue phosphor layer **183** are provided on the ultraviolet light emitting device UV is also possible. In this way, the semiconductor light emitting device can be used in the entire region not only for visible light but also for ultraviolet (UV) light, and can be extended to form a semiconductor light emitting device in which ultraviolet (UV) can be used as an excitation source of the upper phosphor.

Referring back to this example, the semiconductor light emitting device **150** can be positioned on the conductive adhesive layer **130** to constitute a unit pixel in the display device. Since the semiconductor light emitting device **150** has excellent luminance, individual unit pixels can be configured even with a small size. The size of the individual semiconductor light emitting device **150** may have a side length of 80 μm or less, and may be a rectangular or square device. In the case of a rectangle, the size may be $20 \times 80 \mu\text{m}$ or less.

In addition, even when a square semiconductor light emitting device **150** having a side length of 10 μm can be used as a unit pixel, sufficient brightness to form a display device appears. Accordingly, for example, when a unit pixel is a rectangular pixel having one side of 600 μm and the other side of 300 μm , the distance between the semiconductor light emitting devices is relatively large. Accordingly, in this case, it is possible to implement a flexible display device having a high picture quality higher than or equal to HD picture quality.

The display device using the semiconductor light emitting device described above can be manufactured by a new type of manufacturing method. Hereinafter, the manufacturing method will be described with reference to FIG. 6.

FIG. 6 is a cross-sectional view illustrating a method of manufacturing a display device using a semiconductor light emitting device of the present invention.

Referring to this drawing, firstly, the conductive adhesive layer **130** can be formed on the insulating layer **160** in which the auxiliary electrode **170** and the second electrode **140** are positioned. An insulating layer **160** can be laminated on the first substrate **110** to form one substrate wiring substrate), and the wiring substrate may include a first electrode **120**, an auxiliary electrode **170**, and a second electrode **140**. In this case, the first electrode **120** and the second electrode **140** may be disposed in a mutually orthogonal direction. In addition, in order to implement a flexible display device, the first substrate **110** and the insulating layer **160** may each include glass or polyimide (PI).

The conductive adhesive layer **130** may be implemented by, for example, an anisotropic conductive film, and for this purpose, the anisotropic conductive film may be applied to the substrate on which the insulating layer **160** is located.

Next, the second substrate **112** can be disposed to correspond to the positions of the auxiliary electrode **170** and the second electrodes **140**, and a plurality of semiconductors

constituting individual pixels provided on the second substrate **112** may be disposed to face the auxiliary electrode **170** and the second electrode **140**.

In this case, the second substrate **112** is a growth substrate on which the semiconductor light emitting device **150** is grown, and may be a sapphire substrate or a silicon substrate.

When the semiconductor light emitting device is formed in units of wafers, the semiconductor light emitting device can be effectively used in a display device by having an interval and a size that can form a display device.

Next, the wiring board and the second board **112** can be thermocompression-bonded. For example, the wiring substrate and the second substrate **112** may be thermocompression-bonded by applying an ACF press head. The wiring substrate and the second substrate **112** are bonded by the thermocompression bonding. Only a portion between the semiconductor light emitting device **150** and the auxiliary electrode **170** and the second electrode **140** may have conductivity due to the properties of the anisotropic conductive film having conductivity by thermocompression bonding. Accordingly, the electrodes and the semiconductor light emitting device **150** may be electrically connected. At this time, the semiconductor light emitting device **150** can be inserted into the anisotropic conductive film, through which a partition wall may be formed between the semiconductor light emitting devices **150**.

Next, the second substrate **112** can be removed. For example, the second substrate **112** may be removed using a laser lift-off (LLO) method or a chemical lift-off (CLO) method.

Finally, the second substrate **112** can be removed to expose the semiconductor light emitting devices **150** to the outside. If necessary, a transparent insulating layer (not shown) may be formed by coating silicon oxide (SiOx) or the like on the wiring board to which the semiconductor light emitting device **150** is coupled.

The method may further include forming a phosphor layer on one surface of the semiconductor light-emitting device **150**. For example, when the semiconductor light emitting device **150** is a blue semiconductor light emitting device that emits blue (B) light, a red phosphor or a green phosphor may be arranged in a layer form on one surface of the blue semiconductor light emitting device to convert blue (B) light into a color of a unit pixel.

The manufacturing method or structure of the display device using the semiconductor light emitting device described above may be modified in various forms. As an example, a vertical semiconductor light emitting device may also be applied to the display device described above. Hereinafter, a vertical structure will be described with reference to FIGS. 5 and 6.

In addition, in the modification or embodiment described below, the same or similar reference numerals are assigned to the same or similar elements as in the previous example, and the description is replaced with the first description.

FIG. 7 is a perspective view showing another embodiment of a display device using the semiconductor light emitting device of the present invention, FIG. 8 is a cross-sectional view taken along line D-D of FIG. 7, and FIG. 9 is a conceptual diagram showing the vertical semiconductor light emitting device of FIG. 8.

Referring to the drawings, the display device may be a display device using a passive matrix (PM) type vertical semiconductor light emitting device.

The display device may include a substrate **210**, a first electrode **220**, a conductive adhesive layer **230**, a second electrode **240**, and a plurality of semiconductor light emitting devices **250**.

The substrate **210** may be a wiring substrate on which the first electrode **220** is disposed, and may include polyimide (Pt) to implement a flexible display device. In addition, any material that has insulating properties and is flexible may be used.

The first electrode **220** may be positioned on the substrate **210**, and may be formed as a bar-shaped electrode long in one direction. The first electrode **220** may serve as a data electrode.

The conductive adhesive layer **230** may be formed on the substrate **210** on which the first electrode **220** is located. Like a display device to which a flip chip type light emitting device is applied, the conductive adhesive layer **230** may be an anisotropic conductive film (ACF), an anisotropic conductive paste, and a solution containing conductive particles and so on. However, in this embodiment as well, a case in which the conductive adhesive layer **230** is implemented by an anisotropic conductive film will be exemplified.

When the semiconductor light emitting device **250** is connected by applying heat and pressure after the anisotropic conductive film is placed on the substrate **210** in a state where the first electrode **220** is positioned, the semiconductor light emitting device **250** can be electrically connected to the first electrode **220**. In this case, the semiconductor light emitting device **250** is preferably disposed on the first electrode **220**.

As described above, the electrical connection is generated because the anisotropic conductive film has conductivity in the thickness direction when heat and pressure are applied in part. Accordingly, the anisotropic conductive film is divided into a conductive portion and a non-conductive portion in the thickness direction.

In addition, since the anisotropic conductive film contains an adhesive element, the conductive adhesive layer **230** implements not only electrical connection but also mechanical bonding between the semiconductor light emitting device **250** and the first electrode **220**.

As described above, the semiconductor light emitting device **250** is positioned on the conductive adhesive layer **230** and constitutes individual pixels in the display device through this. Since the semiconductor light emitting device **250** has excellent luminance, individual unit pixels can be configured even with a small size. The size of the individual semiconductor light emitting device **250** may have a side length of 80 μm or less, and may be a rectangular or square device. In the case of a rectangular shape, the size may be 20 $\times$ 80 μm or less.

The semiconductor light emitting device **250** may have a vertical structure.

A plurality of second electrodes **240** is disposed between the vertical semiconductor light emitting devices in a direction crossing the longitudinal direction of the first electrode **220** and are provided to be electrically connected the vertical semiconductor light emitting device **250**.

Referring to FIG. 9, such a vertical semiconductor light emitting device may include a p-type electrode **256**, a p-type semiconductor layer **255** formed on the p-type electrode **256**, and a p-type semiconductor layer **255** formed on the p-type semiconductor layer **255**, an active layer **254**, an n-type semiconductor layer **253** formed on the active layer **254**, and an n-type electrode **252** formed on the n-type semiconductor layer **253**. In this case, the lower p-type electrode **256** may be electrically connected to the first

electrode **220** and the conductive adhesive layer **230**, and the upper n-type electrode **252** may be electrically connected to a second electrode **240** to be described later. The vertical semiconductor light emitting device **250** has a great advantage in that it is possible to reduce the chip size because electrodes can be arranged up and down.

Referring back to FIG. 8, a phosphor layer **280** may be formed on one surface of the semiconductor light emitting device **250**. For example, when the semiconductor light emitting device **250** is a blue semiconductor light emitting device **251** that emits blue (B) light, and a phosphor layer **280** for converting the blue (B) light into the color of a unit pixel can be provided. In this case, the phosphor layer **280** may be a red phosphor **281** and a green phosphor **282** constituting individual pixels.

That is, a red phosphor **281** capable of converting blue light into red (R) light may be stacked on the blue semiconductor light emitting device at a position constituting the red unit pixel. In addition, a green phosphor **282** capable of converting blue light into green (G) light may be stacked on the blue semiconductor light emitting device at a position forming the green unit pixel. In addition, only the blue semiconductor light emitting device may be used alone in the portion constituting the blue unit pixel. In this case, unit pixels of red (R), green (G), and blue (B) may form one pixel.

However, the present invention is not necessarily limited thereto, and as described above in a display device to which a flip chip type light emitting device is applied, other structures for realizing blue, red, and green may be applied.

Referring to this embodiment again, the second electrode **240** may be positioned between the semiconductor light emitting devices **250** and may be electrically connected to the semiconductor light emitting devices **250**. For example, the semiconductor light emitting devices **250** may be arranged in a plurality of columns, and the second electrode **240** may be positioned between the columns of the semiconductor light emitting devices **250**.

Since the distance between the semiconductor light emitting devices **250** constituting individual pixels is sufficiently large, the second electrode **240** may be positioned between the semiconductor light emitting devices **250**.

The second electrode **240** may be formed as a long bar-shaped electrode in one direction, and may be disposed in a direction perpendicular to the first electrode.

In addition, the second electrode **240** and the semiconductor light emitting device **250** may be electrically connected by a connection electrode protruding from the second electrode **240**. More specifically, the connection electrode may be an n-type electrode of the semiconductor light emitting device **250**. For example, the n-type electrode is formed as an ohmic electrode for ohmic contact, and the second electrode covers at least a portion of the ohmic electrode by printing or deposition. Through this, the second electrode **240** and the n-type electrode of the semiconductor light emitting device **250** may be electrically connected.

As shown in drawings, the second electrode **240** may be positioned on the conductive adhesive layer **230**. In some cases, a transparent insulating layer (not shown) including silicon oxide (SiOx) may be formed on the substrate **210** on which the semiconductor light emitting device **250** is formed. When the second electrode **240** is positioned after the transparent insulating layer is formed, the second electrode **240** is positioned on the transparent insulating layer. In addition, the second electrode **240** may be formed to be spaced apart from the conductive adhesive layer **230** or the transparent insulating layer.

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If a transparent electrode such as indium tin oxide (ITO) is used to position the second electrode **240** on the semiconductor light emitting device **250**, there is a problem in that the ITO material has bad adhesion to the n-type semiconductor layer. Accordingly, the present invention has the advantage of not using a transparent electrode such as ITO by locating the second electrode **240** between the semiconductor light emitting devices **250**. Therefore, it is possible to improve light extraction efficiency by using a conductive material having good adhesion to the n-type semiconductor layer as a horizontal electrode without being limited by the selection of a transparent material.

According to the drawings, a partition wall **290** may be positioned between the semiconductor light emitting devices **250**. That is, in order to isolate the semiconductor light emitting devices **250** constituting individual pixels, a partition wall **290** may be disposed between the vertical semiconductor light emitting devices **250**. In this case, the partition wall **290** may serve to separate individual unit pixels from each other, and may be integrally formed with the conductive adhesive layer **230**. For example, by inserting the semiconductor light emitting device **250** into the anisotropic conductive film, the base member of the anisotropic conductive film may form the partition wall.

In addition, when the base member of the anisotropic conductive film is black, the partition wall **290** may have reflective properties and increase contrast even without a separate black insulator.

As another example, as the partition wall **190**, a reflective partition wall may be separately provided. The partition wall **290** may include a black or white insulator depending on the purpose of the display device.

If the second electrode **240** is directly positioned on the conductive adhesive layer **230** between the semiconductor light emitting device **250**, the partition wall **290** can be disposed between the vertical semiconductor light emitting device **250** and the second electrode **240**. Accordingly, individual unit pixels can be configured with a small size by using the semiconductor light emitting device **250**, and the distance between the semiconductor light emitting devices **250** is relatively large enough to connect the second electrode **240** to the semiconductor light emitting device **250**, and there is an effect of realizing a flexible display device having HD image quality.

Also, as illustrated, a black matrix **291** may be disposed between each phosphor to improve contrast. That is, the black matrix **291** may improve contrast of light and dark.

As described above, the semiconductor light emitting device **250** can be positioned on the conductive adhesive layer **230** and constitutes individual pixels in the display device. Since the semiconductor light emitting device **250** has excellent luminance, individual unit pixels can be configured even with a small size. Accordingly, a full-color display in which unit pixels of red (R), green (G), and blue (B) constitute one pixel can be implemented by the semiconductor light emitting device.

FIG. **10** is a diagram schematically illustrating an embodiment of a method for assembling a semiconductor light emitting device to a display panel by a self-assembly method, and FIG. **11** is an enlarged view of part A of FIG. **10**.

An example in which a semiconductor light emitting device is assembled in a display panel by a self-assembly method using an electromagnetic field will be briefly described with reference to FIGS. **10** to **11**.

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Referring to FIGS. **10** to **11**, the semiconductor light emitting device **1100** may be introduced into a chamber **1300** filled with a fluid **1200**.

The semiconductor light emitting device **1100** may be implemented as the horizontal semiconductor light emitting device shown in FIG. **4** or the vertical semiconductor light emitting device shown in FIG. **9**. In addition, the semiconductor light emitting device **1100** may include a magnetic layer **1150** having a magnetic material. The magnetic layer may include a magnetic metal such as nickel (Ni). Since the semiconductor light emitting device **1100** injected into the fluid includes the magnetic layer **1150**, it moves to the display panel **1000** (hereinafter defined as 'panel') by the magnetic field generated from the assembly apparatus **1400** and can be assembled to the panel **1000**.

Referring to the flipped semiconductor light emitting device **1100** shown in FIG. **11**, the magnetic layer **1150** of the semiconductor light emitting device **1100** is shown as being disposed on the p-type semiconductor layer **1130**, but the arrangement position of the magnetic layer **1150** may be changed according to the manufacturing method of the light emitting device **1100**.

Meanwhile, a passivation layer **1170** surrounding the top and side surfaces of the semiconductor light emitting device **1100** may be formed on the semiconductor light emitting device **1100**. Since the passivation layer **1170** is formed while the semiconductor light emitting device **1100** is connected to a growth substrate (not shown), the passivation layer **1170** may not be formed on the bottom surface of the semiconductor light emitting device **1100**.

The passivation layer **1170** can be formed through an inorganic insulator such as silica, alumina, PECVD (Plasma Enhanced Chemical Vapor Deposition), LPCVD (Low Pressure CVD), sputtering deposition method, or the like, or it may be formed through a method of spin coating an organic material such as or a photoresist, a polymer material.

Meanwhile, an n-type electrode may be connected to the n-type semiconductor layer **1110**, and a p-type electrode may be connected to the p-type semiconductor layer **1130**. To this end, some regions of the n-type semiconductor layer **1110** and the p-type semiconductor layer **1130** is exposed to the outside. Accordingly, in the manufacturing process of the display device after the semiconductor light emitting device **1100** is assembled to the panel **1000**, a partial region of the passivation layer **1170** may be etched.

After the semiconductor light emitting device **1100** is put into the chamber **1300**, the panel **1000** may be disposed on the chamber **1300**. According to an embodiment, the panel **1000** may be introduced into the chamber **1300**.

A pair of assembly electrodes **1011** and **1012** corresponding to each of the semiconductor light emitting devices **1100** to be assembled may be formed on the panel **1000**. The assembly electrodes **1011** and **1012** may be implemented as transparent electrodes (ITO) or other common materials. The assembly electrodes **1011** and **1012** may fix the semiconductor light emitting device **1100** injected into the coupling hole **1001** by emitting an electric field as a voltage is applied thereto. The distance between the assembly electrodes **1011** and **1012** is formed to be smaller than the width of the semiconductor light emitting device **1100** and the width of the coupling hole **1001**, so that the assembly position of the semiconductor light emitting device **1100** using an electric field can be more precisely fixed.

An insulating layer **1020** can be formed on the assembly electrodes **1011** and **1012** to protect the assembly electrodes **1011** and **1012** from the fluid **1200**, and to prevent leakage of current flowing through the assembly electrodes **1011** and

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1012. For example, the insulating layer 1020 may be formed of a single layer or multiple layers of an inorganic insulator such as silica or alumina or an organic insulator. The insulating layer 1020 may have a minimum thickness for preventing damage to the assembly electrodes 1011 and 1012 when assembling the semiconductor light emitting device 1100, and may have a maximum thickness for stably assembling the semiconductor light emitting device 1100.

A partition wall 1030 may be formed on the insulating layer 1020. Partial regions of the partition wall 1030 may be positioned on the assembly electrodes 1011 and 1012, and the remaining regions may be positioned on the panel 1000.

For example, when the panel 1000 is manufactured, a portion of the partition wall formed on the entire upper portion of the insulating layer 1020 is removed, a coupling hole 1001 can be formed such that each of the semiconductor light emitting devices 1100 is coupled and assembled to the panel 1000.

A coupling hole 1001 to which the semiconductor light emitting devices 1100 are coupled is formed in the panel 1000, and a surface on which the coupling hole 1001 is formed may be in contact with the fluid 1200. The coupling hole 1001 may guide an accurate assembly position of the semiconductor light emitting device 1100.

Meanwhile, the coupling hole 1001 may have a shape and a size corresponding to the shape of the semiconductor light emitting device 1100 to be assembled at a corresponding position. Accordingly, it is possible to prevent assembling other semiconductor light emitting devices or assembling a plurality of semiconductor light emitting devices in the coupling hole 1001.

After the panel 1000 is disposed, the assembly apparatus 1400 including the magnetic material may move along the panel 1000. The assembly apparatus 1400 may move while in contact with the panel 1000 in order to maximize the area applied by the magnetic field into the fluid 1200. According to an embodiment, the assembly apparatus 1400 may include a plurality of magnetic materials or a magnetic material having a size corresponding to that of the panel 1000. In this case, the moving distance of the assembly apparatus 1400 may be limited within a predetermined range.

By the magnetic field generated by the assembly apparatus 1400, the semiconductor light emitting device 1100 in the chamber 1300 may move toward the assembly apparatus 1400 and the panel 1000.

The semiconductor light emitting device 1100 may enter the coupling hole 1001 and come into contact with the panel 1000 while moving toward the assembly apparatus 1400. For example, a pattern or shape for the n-type semiconductor layer 1110 of the semiconductor light emitting device 1100 to contact the panel 1000 may be formed in the coupling hole 1001 and/or the semiconductor light emitting device 1100.

In addition, the electric field by the assembly electrodes 1011 and 1012 formed on the panel 1000 can prevent the semiconductor light emitting device 1100 in contact with the panel 1000 from being separated by the movement of the assembly apparatus 1400.

That is, by the self-assembly method using the above-described electromagnetic field, the time required for each of the semiconductor light emitting devices to be assembled on the substrate can be rapidly reduced, so that a large-area high-pixel display can be implemented more quickly and economically.

However, in the case of the self-assembly method, the assembly electrodes 1011 and 1012 for assembling the

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semiconductor light emitting device 1100 to the panel 1000 by applying an electric field to the panel 1000 are needed. In addition, a separate wiring for applying a voltage to the assembly electrodes 1011 and 1012 may be required in the panel 1000. In this case, an additional process is required to implement the separate wiring. Also, depending on the implementation of the wiring, noise may be generated in other circuits or a reduction of space utilization in display pixels may occur.

A display device according to an embodiment of the present invention for solving the above-described problem will be described with reference to FIGS. 12 to 18 below.

FIG. 12 is a diagram schematically showing a circuit configuration of a display panel according to an embodiment of the present invention.

Referring to FIG. 12, a driving circuit for controlling each of the pixels implemented by the semiconductor light emitting devices 1100 may be implemented in the panel 1000. For example, the driving circuit may be implemented as a thin film transistor (TFT) driving circuit.

The TFT driving circuit is a known configuration, for example, the TFT driving circuit may include a power supply wiring (or a drain wiring VDD 1041) supplying a voltage for one frame, a ground wiring (or a source wiring VSS 1042), a capacitor 1070 storing a voltage for a pixel, a driving TFT 1050 connected to the capacitor 1070 by the wiring 1072 and controlling an amount of current flowing to the semiconductor light emitting device 1100 based on the voltage stored in the capacitor 1070, a switch TFT 1080 controlling the light of the entire pixel, a data line VDATA 1091 for applying a voltage to the pixel, and a gate line 1092 for controlling on/off of the switch TFT 1080.

The power wiring 1041 may be connected to the first electrode 1112 (refer to FIG. 15, for example, an n-type electrode) of the semiconductor light emitting device 1100 through a first contact 1063. The drain terminal 1051 of the driving TFT 1050 may be connected to the second electrode 1132 (refer to FIG. 15, for example, a p-type electrode) of the semiconductor light emitting device 1100 through a second contact 1064. Also, the source terminal 1052 of the driving TFT 1050 may be connected to the ground wire 1042 through the third contact 1066. As will be described later with reference to FIGS. 15 to 18, the first contact 1063 and the second contact 1064 may mean a first trench region and a second trench region which are formed by etching at least a portion of the partition wall 1030 and the insulating layer 1020.

On the other hand, the first assembly electrode 1011 and the second assembly electrode 1012 can be arranged such that a partial region overlaps with the semiconductor light emitting device 1100 when viewed from above, so that an electric field to fix the semiconductor light emitting device 1100 on a panel 000 can be effectively applied. In particular, one end of the first assembly electrode 1011 may be connected to the power wiring 1041, and one end of the second assembly electrode 1012 may be connected to the ground wiring 1042. In this case, in the assembly process of the semiconductor light emitting device 1100, as a predetermined voltage can be applied to the first assembly electrode 1011 through the power supply wiring 1041, an electric field can develop between the first assembly electrode 1011 and the second assembly electrode 1012.

When the first assembly electrode 1011 is connected to any one of the other wirings 1091, 1092, and 1072 shown in FIG. 12, in order to apply a voltage to the first assembly electrode 1011, a plurality of wirings voltage are applied to them. That is, according to the embodiment of the present invention, the

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first assembly electrode **1011** may be connected to the power wiring **1041** to more easily generate an electric field during the assembly process of the semiconductor light emitting device **1100**.

FIGS. **13** to **14** are circuit diagrams according to the circuit configuration of the display panel shown in FIG. **12**.

Referring to FIG. **13**, during the assembly process of the semiconductor light emitting device **1100**, when a voltage is applied to the first assembly electrode **1011** having one end connected to the power wiring **1041**, an electric field may be generated between the first assembly electrode **1011** and the second assembly electrode **1012**. The semiconductor light emitting device **1100** accommodated in the coupling hole **1001** of the panel **1000** by the magnetic field of the assembly apparatus **1400** may maintain a state fixed to the panel **1000** by the electric field.

Referring to FIG. **14**, after the semiconductor light emitting device **1100** is assembled, a wiring connection process may be performed. For example, the first wiring **1061** may be connected between the first electrode **1112** (refer to FIG. **15**) of the semiconductor light emitting device **1100** and the power wiring **1041**. Also, a second wiring **1062** may be connected between the second electrode **1132** (refer to FIG. **15**) of the semiconductor light emitting device **1100** and the drain terminal **1051** of the driving TFT **1050**. In this case, the other end of the first assembly electrode **1011** or the second assembly electrode **1012** may not be connected to the first wiring **1061** or the second wiring **1062**. Accordingly, the first assembly electrode **1011** and the second assembly electrode **1012** may not affect the semiconductor light emitting device **1100** and the TFT driving circuit when the manufactured display device is driven. Meanwhile, the first wiring **1061** and the second wiring **1062** may be implemented as a transparent electrode (ITO).

That is, according to the embodiment of the present invention, the assembly electrodes **1011** and **1012** for self-assembly of the semiconductor light emitting device **1100** are connected to the power supply wiring **1041** and the ground wiring **1042** of the TFT driving circuit such that a separate power wiring for applying a voltage to the assembly electrodes **1011** and **1012** may not be implemented in the panel **1100**. Accordingly, it is possible to prevent the wiring structure of the panel **1100** from becoming complicated, reduce noise due to wiring, and prevent a decrease in space utilization within the pixel.

FIGS. **15** to **18** are exemplary views showing various arrangement positions of the assembly electrode according to an embodiment of the present invention.

Referring to FIGS. **15** to **18**, assembly electrodes **1011** and **1012** may be formed in various positions of the panel **1100** to be connected to a power supply wiring **1041** and a ground wiring **1042**.

First, referring to the embodiment of FIG. **15**, on the substrate **1002** of the panel **1100**, the power wiring **1041** and the ground wiring **1042** may be disposed to be spaced apart from each other. Also, the gate terminal **1053** of the driving TFT **1050** may be formed on the substrate **1002**.

Thereafter, a first insulating layer **1021** covering the upper portion of the substrate **1002**, the power wiring **1041**, the ground wiring **1042**, and the gate terminal **1053** may be formed.

A drain terminal **1051** and a source terminal **1052** of the driving TFT **1050** may be formed on the first insulating layer **1021**. In particular, the source terminal **1052** can be formed after a partial region on the ground wire **1042** of the first

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insulating layer **1021** is etched, so that the source terminal **1052** and the ground wire **1042** may be electrically connected to each other.

Also, a partial region of the first insulating layer **1021** present on the power wiring **1041** may be etched, and a terminal electrically connected to the power wiring **1041** may be formed in the etched region.

Thereafter, a second insulating layer **1022** and a third insulating layer **1023** covering the first insulating layer **1021**, the drain terminal **1051**, and the source terminal **1052** may be formed. Each of the second insulating layer **1022** and the third insulating layer **1023** may be implemented with the same material as the first insulating layer **1021**, or may be implemented with various known materials used for the insulating layer.

According to an embodiment, a gate terminal **1052** may be formed over a region between the drain terminal **1051** and the source terminal **1052** of the third insulating layer **1023**. In this case, the gate terminal **1052** may not be formed on the substrate **1002**.

On the other hand, some regions of the second insulating layer **1022** and the third insulating layer **1023** present on the power wiring **1041** and the ground wiring **1042** are etched, and a terminal electrically connected to the power wiring **1041** can be formed in the etched region and a terminal electrically connected to the source terminal **1052** may be formed.

Thereafter, a fourth insulating layer **1024** may be formed on the third insulating layer **1023**. The fourth insulating layer **1024** may be made of the same material as at least one of the first insulating layer **1021** to the third insulating layer **1023**, or a material used for the insulating layer.

A first assembly electrode **1011** and a second assembly electrode **1012** may be formed on the fourth insulating layer **1024**. On the other hand, some regions of the fourth insulating layer **1024** present in the power wiring **1041** and the ground wiring **1042** are etched, and the first assembly electrode **1011** can be electrically connected to the power wiring **1041** through the etched region. The second assembly electrode **1012** may be formed to be electrically connected to the ground wiring **1042** through the etched region.

Thereafter, the fifth insulating layer **1025** and the partition wall **1030** may be formed on the fourth insulating layer **1024**. In addition, a region in which the semiconductor light emitting device **1100** is to be assembled among the partition walls **1030** may be etched to form a coupling hole **1001**.

The panel **1000** in which the coupling hole **1001** is formed is disposed on the chamber **1300** in which the fluid **1200** is accommodated, as shown in FIGS. **10** to **11**, and the semiconductor light emitting device **1100** may be inserted into the coupling hole **1001** and assembled to the panel **1000**. As described above, since the insulating layer **1025** is formed on the assembly electrodes **1011** and **1012**, contact between the fluid **1200** and the assembly electrodes **1011** and **1012** may be prevented.

After the semiconductor light emitting device **1100** is assembled to the panel **1000**, an electrode formation process of the semiconductor light emitting device **1100** may be performed. For example, a portion of the passivation layer **1170** and a portion of the second semiconductor layer **1130** of the semiconductor light emitting device **1100** can be etched to expose the first semiconductor layer **1110** and the second semiconductor layer **1130** to the outside. In addition, a first electrode **1112** may be formed on the exposed first semiconductor layer **1110**, and a second electrode **1132** may be formed on the second semiconductor layer **1130**.

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After the electrode formation process is performed, a portion of the partition wall **1030** and the fifth insulating layer **1025** on the power wiring **1041** may be etched to form a first trench region (or a first contact **1063**). A first wiring **1061** connecting the first assembly electrode **1011** exposed through the first trench region **1063** and the first electrode **1112** may be formed. The first wiring **1061** may be electrically connected to the wiring electrode **1041** as it is connected to the first assembly electrode **1011**.

Further, a second trench region (or a second contact **1064**) may be formed by etching a partial region of the partition wall **1030**, the second insulating layer **1022** to the fifth insulating layer **1025** above the drain terminal **1051** of the driving TFT **1050**. Thereafter, a second wiring **1064** connecting the drain terminal **1051** exposed through the second trench region **1064** and the second electrode **1132** may be formed.

In the case of the embodiment shown in FIG. **15**, since the assembly electrodes **1011** and **1012** may be formed at the position closest to the semiconductor light emitting device **1100**, an electric field can be more effectively applied to the semiconductor light emitting device **1100**. Accordingly, the assembly rate of the semiconductor light emitting device **1100** may be maximized.

Meanwhile, according to the embodiment of FIG. **15**, a plurality of etching processes could be needed to connect the assembly electrodes **1011** and **1012** to the power wiring **1041** and the ground wiring **1042**.

Based on this, referring to FIG. **16**, the assembly electrodes **1011** and **1012** may be formed on the third insulating layer **1023**. In this case, since the number of etching processes for connecting the assembly electrodes **1011** and **1012** to the power wiring **1041** and the ground wiring **1042** can be reduced compared to FIG. **15**, the process for forming the assembly electrode may be more efficient.

Also, referring to the embodiment of FIG. **17**, the assembly electrodes **1011** and **1012** may be formed on the first insulating layer **1021**. In this case, since an additional etching process for connecting the assembly electrodes **1011** and **1012** to the power wiring **1041** and the ground wiring **1042** is unnecessary, the process for forming the assembly electrode can be reduced compared to the embodiment of FIGS. **15** and **16** and can be more efficient. In particular, according to the embodiment of FIG. **17**, since the second assembly electrode **1012** may be formed to extend from the source terminal **1052**, the source terminal **1052** and the second assembly electrode **1012** can be formed in one process.

Also, referring to the embodiment of FIG. **18**, the assembly electrodes **1011** and **1012** may be directly formed on the substrate **1002**. In this case, as in the embodiment of FIG. **17**, an etching process for connecting the assembly electrodes **1011** and **1012** to the power wiring **1041** and the ground wiring **1042** is unnecessary. In particular, according to the embodiment of FIG. **18**, the first assembly electrode **1011** may be formed to extend from the power wiring **1041**, and the second assembly electrode **1012** may be formed to extend from the ground wiring **1042**. That is, the first assembly electrode **1011** and the power wiring **1041** can be formed in one process, and the second assembly electrode **1012** and the ground wiring **1042** can be formed in a single process such that forming process of assembly electrode can be done most efficiently.

The above description is merely illustrative of the technical idea of the present invention, and various modifications and variations are possible without departing from the

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essential characteristics of the present invention by those skilled in the art to which the present invention pertains.

Accordingly, the embodiments disclosed in the present invention are not intended to limit the technical spirit of the present invention, but to explain, and the scope of the technical spirit of the present invention is not limited by these embodiments.

The protection scope of the present invention should be interpreted by the following claims, and all technical ideas within the scope equivalent thereto should be construed as being included in the scope of the present invention.

The invention claimed is:

1. A display device comprising;

a substrate;

a power wiring and a ground wiring disposed on the substrate and spaced apart from each other;

a driving thin film transistor (TFT) disposed on the substrate and having a source terminal electrically connected to the ground wiring;

at least one insulating layer disposed on the substrate;

a pair of assembly electrodes including a first assembly electrode and a second assembly electrode spaced apart from each other and disposed between the at least one insulating layer and the substrate;

a semiconductor light emitting device including a first electrode and a second electrode, and assembled on the at least one insulating layer;

a first wiring electrically connecting the first electrode of the semiconductor light emitting device, the first assembly electrode, and the power wiring;

a second wiring electrically connecting the second electrode of the semiconductor light emitting device and a drain terminal of the driving TFT; and

a partition wall including a first trench region on the at least one insulating layer, the partition wall further comprising a coupling hole that accommodates the semiconductor light emitting device,

wherein the pair of assembly electrodes is configured to generate an electric field as a voltage is applied to any one of the pair of assembly electrodes,

wherein the first wiring connects the first electrode and the first assembly electrode via the first trench region disposed in the partition wall,

wherein the second assembly electrode separated from the second wiring,

wherein an uppermost surface of the partition wall and an upper surface of the semiconductor light emitting device contacting the first wiring are coplanar, and

wherein a bottommost surface of the partition wall and a bottommost surface of the semiconductor light emitting device are coplanar.

2. The display device of claim 1, wherein the second assembly electrode of the pair of assembly electrodes is electrically connected to the ground wiring.

3. The display device according to claim 2, wherein the pair of assembly electrodes are formed on the substrate, the first assembly electrode is formed to extend from the power wiring, and the second assembly electrode is formed to extend from the ground wiring.

4. The display device according to claim 2, wherein the at least one insulating layer comprises a first insulating layer covering the substrate, the power wiring, and the ground wiring,

wherein the first assembly electrode, the second assembly electrode, and the source terminal are formed on the first insulating layer,

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wherein the first assembly electrode is electrically connected to the power wiring through an etching region formed on the power wiring of the first insulating layer, and

wherein the second assembly electrode is formed to extend from the source terminal and is electrically connected to the ground wire through an etch region formed on the ground wire in the first insulating layer.

5 The display device according to claim 2, wherein the at least one insulating layer comprises:

- a first insulating layer covering the substrate, the power wiring, and the ground wiring;
- a second insulating layer covering the source terminal formed on the first insulating layer and covering the first insulating layer; and
- a third insulating layer covering the first assembly electrode and the second assembly electrode formed on the second insulating layer, and the second insulating layer;

wherein the first assembly electrode is electrically connected to the power wiring through an etch region formed in each of the first insulating layer and the second insulating layer on the power wiring, and

wherein the second assembly electrode is electrically connected to the source terminal and the ground wire through an etch region formed in the second insulating layer on the ground wire.

6 The display device according to claim 2, wherein the at least one insulating layer comprises:

- a first insulating layer covering the substrate, the power wiring, and the ground wiring;
- a second insulating layer covering the source terminal formed on the first insulating layer and covering the first insulating layer; and
- a third insulating layer on the second insulating layer, and

wherein the first assembly electrode and the second assembly electrode are disposed on the third insulating layer.

7 The display device according to claim 2, wherein the at least one insulating layer comprises:

- a first insulating layer covering the substrate, the power wiring, and the ground wiring;
- a second insulating layer covering the source terminal formed on the first insulating layer and covering the first insulating layer; and
- a third insulating layer on the second insulating layer, and

wherein the first assembly electrode and the second assembly electrode are disposed on the first insulating layer.

8 The display device according to claim 7, wherein the second assembly electrode is extended from the source terminal, and the source terminal and the second assembly electrode are formed into one body.

9 The display device according to claim 2, wherein the at least one insulating layer comprises:

- a first insulating layer covering the substrate, the power wiring, and the ground wiring;
- a second insulating layer covering the source terminal formed on the first insulating layer and covering the first insulating layer; and
- a third insulating layer on the second insulating layer, and

wherein the first assembly electrode and the second assembly electrode are directly disposed on the substrate.

10 The display device according to claim 9, wherein the first assembly electrode is extended from the power wiring, and the first assembly electrode and the power wiring are formed into one body.

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11 The display device according to claim 9, wherein the second assembly electrode is extended from the ground wiring, and the second assembly electrode and the ground wiring are formed into one body.

12 The display device according to claim 1, wherein the semiconductor light emitting device is positioned on the pair of assembly electrodes.

13 The display device according to claim 12, wherein the semiconductor light emitting device includes a magnetic layer having a magnetic material.

14 The display device according to claim 1, wherein the first wiring is electrically connected to the power wiring through the partition wall on the power wiring and the first trench region provided by etching the at least one insulating layer, and

wherein the second wiring is electrically connected to the drain terminal through the partition wall on the drain terminal and a second trench region provided by etching the at least one insulating layer.

15 The display device according to claim 1, wherein each of the first wiring and the second wiring is implemented as a transparent electrode.

16 The display device according to claim 1, wherein a spaced distance between the first assembly electrode and the second assembly electrode is less than a lateral width of the semiconductor light emitting device.

17 The display device according to claim 1, wherein the partition wall includes a second trench region exposing the drain terminal of the driving TFT to connect the second electrode and the drain terminal of the driving TFT.

18 A display device comprising;

- a power wiring and a ground wiring disposed on a substrate and spaced apart from each other;
- a driving thin film transistor (TFT) disposed on the substrate and having a source terminal electrically connected to the ground wiring;
- at least one insulating layer disposed on the substrate;
- a light emitting device including a first electrode and a second electrode on the substrate;
- a pair of assembly electrodes disposed between the at least one insulating layer and the substrate, and including a first assembly electrode and a second assembly electrode spaced apart from each other;
- a first wiring electrically connecting the first electrode of the light emitting device, the first assembly electrode, and the power wiring;
- a second wiring electrically connecting the second electrode of the light emitting device and a drain terminal of the driving TFT; and
- a partition wall including a first trench region on the at least one insulating layer, the partition wall further comprising a coupling hole that accommodates the light emitting device,

wherein the light emitting device is separated from the pair of assembly electrodes by the at least one insulating layer,

wherein the pair of assembly electrodes is configured to generate an electric field as a voltage is applied to any one of the pair of assembly electrodes,

wherein the first wiring connects the first electrode and the first assembly electrode via the first trench region disposed in the partition wall,

wherein the second assembly electrode is separated from the second wiring,

wherein an uppermost surface of the partition wall and an upper surface of the light emitting device contacting the first wiring are coplanar, and

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wherein a bottommost surface of the partition wall and a bottommost surface of the semiconductor light emitting device are coplanar.

19. The display device according to claim 18, wherein the at least one insulating layer is interposed between the driving TFT and the light emitting device. 5

20. The display device according to claim 18, wherein the first wiring and the second wiring penetrate through the at least one insulating layer.

21. The display device according to claim 18, wherein both the first assembly electrode and the second assembly electrode are vertically overlapped with the first electrode of the light emitting device. 10

22. A display device comprising;

a substrate;

a power wiring disposed on the substrate;

at least one insulating layer disposed on the substrate;

a pair of assembly electrodes including a first assembly electrode and a second assembly electrode spaced apart from each other and disposed between the at least one insulating layer and the substrate; 15 20

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a semiconductor light emitting device including an electrode assembled on the at least one insulating layer;

a wiring electrically connecting the electrode of the semiconductor light emitting device, the first assembly electrode, and the power wiring; and

a partition wall including a trench hole on the at least one insulating layer, the partition wall further comprising a coupling hole that accommodates the semiconductor light emitting device,

wherein the pair of assembly electrodes is configured to generate an electric field as a voltage is applied to any one of the pair of assembly electrodes,

wherein the wiring connects the electrode, the first assembly electrode, and the power wiring via the trench hole disposed in the partition wall,

wherein an uppermost surface of the partition wall and an upper surface of the semiconductor light emitting device contacting the wiring are coplanar, and

wherein a bottommost surface of the partition wall and a bottommost surface of the semiconductor light emitting device are coplanar. 25

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